

TrES-5b

R-2010

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_150808 · created 2026-08-07T17:02:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457242.7033 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.054
Transit depth (Rp/Rs)^2	3.2 +/- 1.94 [%]
Orbital Inclination (inc)	84.42 +/- 2.9
Airmass coefficient 1 (a1)	0.045 +/- 0.038
Airmass coefficient 2 (a2)	3.0 +/- 0.68
Scatter in the residuals of the lightcurve fit is	134.27 %
Best Comparison Star	None
Optimal Aperture	4.7
Optimal Annulus	7.05
Transit Duration (day)	0.072 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.179 ± 0.054 vs 0.142 (deviation 0.44σ)
Duration fit vs published	0.072 d vs 0.0758 d (deviation 0.12σ)
Mid-time O–C	3.8 min
Depth SNR	2.1
Residual scatter	134.27 %

Light curve

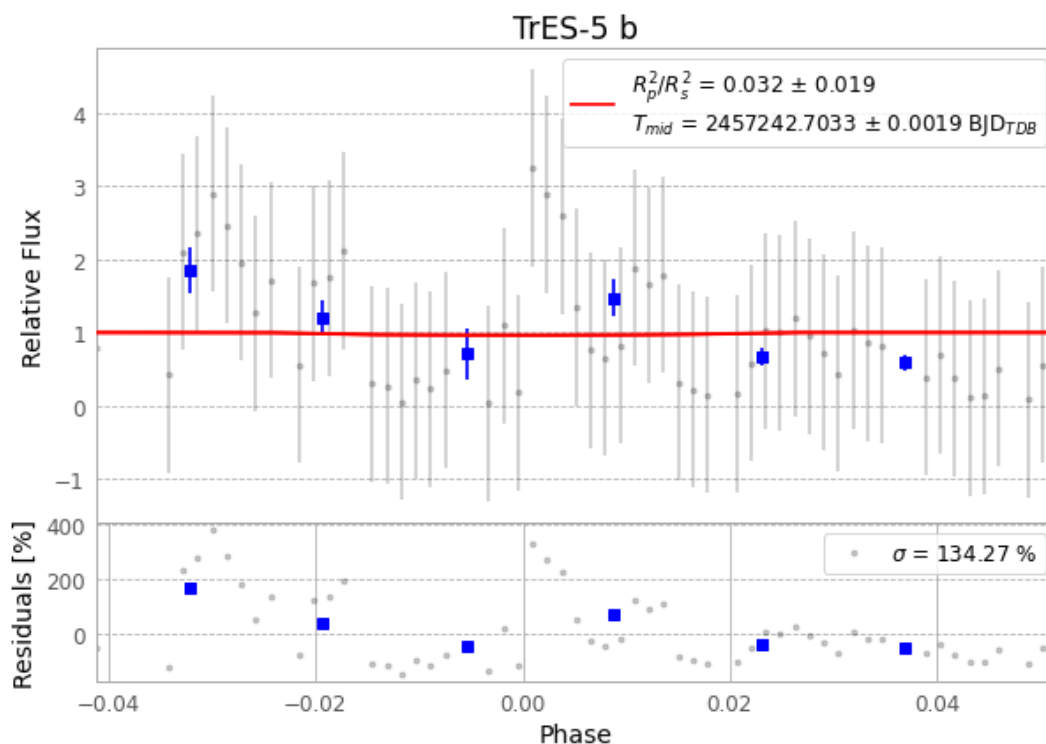


Figure 1 — FinalLightCurve_TrES-5 b_2015-08-07.png (SHA-256 dbf79228cd413198...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [204, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0bea580076a27738...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

65 FITS frames (43 MB), TRES-5150808032115.FITS → TRES-5150808063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-150808.log (c845fdadc8c8...), FinalLightCurve_TrES-5 b_2015-08-07.png (dbf79228cd41...), FinalLightCurve_TrES-5 b_2015-08-07.csv (dddc1f4e4f1c...)

Compute resources

Wall time 179.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 17:02Z.